

CHAPTER – 2

Zara-Fashion-E-commerce management

Identify entities, attributes, primary keys, and relationships required for the system

2.Introduction

Identifying entities, attributes, primary keys, and relationships is an essential step in designing a database for an e-commerce system. Entities represent the main objects in the system, attributes describe the properties of those entities, primary keys uniquely identify each record, and relationships define how different entities are connected. This structured approach ensures efficient data organization, maintains data integrity, and supports the smooth functioning of the Zara-Inspired Fashion E-Commerce System.

2.1 Customer

Attribute	Description	Key
Customer_ID	Unique identifier for each customer	Primary Key
First_Name	Customer's first name	Attribute
Email	Customer's email address	Attribute
Phone_Number	Customer's contact number	Attribute
Address	Customer's residential address	Attribute

2.2 Category

Attribute	Description	Key
Category_ID	Unique identifier for each category	Primary Key
Category_Name	Name of the product category	Attribute
Description	Details about the category	Attribute

2.3 Product

Attribute	Description	Key
Product_ID	Unique identifier for each product	Primary Key
Product_Name	Name of the product	Attribute
Brand	Brand of the product	Attribute
Price	Selling price of the product	Attribute
Size	Available size of the product	Attribute
Colour	Available colour of the product	Attribute
Category_ID	Links the product to a category	Foreign Key

2.4 Cart

Attribute	Description	Key
Cart_ID	Unique identifier for each shopping cart	Primary Key
Customer_ID	Identifies the customer who owns the cart	Foreign Key
Product_ID	Identifies the product added to the cart	Foreign Key
Quantity	Identifies the product added to the cart	Attribute

2.5 Order

Attribute	Description	Key
Order_ID	Unique identifier for each order	Primary Key
Customer_ID	Identifies the customer who placed the order	Foreign Key
Order_Date	Date on which the order was placed	Attribute
Total_Amount	Total amount of the order	Attribute
Order_Status	Current status of the order	Attribute

2.6 Order Item

Attribute	Description	Key
Order_Item_ID	Unique identifier for each order item	Primary Key
Order_ID	Identifies the associated order	Foreign Key
Product_ID	Identifies the ordered product	Foreign Key
Quantity	Number of products ordered	Attribute
Price	Price of the ordered product	Attribute

2.7 Payment

Attribute	Description	Key
Payment_ID	Unique identifier for each payment	Primary Key
Order_ID	Identifies the related order	Foreign Key
Payment_Method	Method used for payment	Attribute
Payment_Date	Date of payment	Attribute
Payment_Status	Current payment status	Attribute
Amount	Total payment amount	Attribute

2.8 Relationship

Entity	Relationship	Related Entity	Cardinality
Customer	Places	Order	One Customer can place Many Orders (1:M)
Customer	Owns	Cart	One Customer has One Shopping Cart (1:1)
Category	Contains	Product	One Category can contain Many Products (1:M)
Product	Belongs To	Category	Many Products belong to One Category (M:1)

Cart	Contains	Product	One Cart can contain Many Products, and One Product can be added to Many Carts (M:N)
Order	Contains	Order_Item	One Order can contain Many Order Items (1:M)
Product	Included In	Order_Item	One Product can appear in Many Order Items (1:M)
Order	Has	Payment	One Order has One Payment (1:1)
Customer	Makes	Payment	One Customer can make Many Payments (1:M)

2.9 Conclusion

A well-designed database structure is essential for the successful implementation of an e-commerce system. By clearly defining the entities, their attributes, primary keys, and relationships, the Zara-Inspired Fashion E-Commerce System ensures accurate data management and efficient communication between different modules.